

Scientific Computing 2026

Week 3: September 14, 16

September 14, 2026

Monday: Eigensystems

Topics: Eigensystems: What are they and how do they work
Algebraic vs geometric multiplicity: defective vs non-defective systems
Transform to basis of eigenvectors
The power of diagonalization: Continuous functions of bounded linear operators
Solving eigensystems solves linear eqs. (including singular)
Simplest method: Power iteration

Read beforehand: Chapter 4.1, 4.2, 4.4, 4.5.1

Wednesday: Iterative Eigensolvers

Topics: Inverse power iteration
Locating eigenvalues: Gershgorin's Theorem
Zooming in on eigenvalues: Shifted inverse iteration
Superlinear convergence: Raleigh iteration
Krylov subspace methods: Arnoldi and Lanczos

Read beforehand: Chapter 4.5.2–4.5.7

Review Questions: 4.1–4.7, 4.10–4.12

Exercises: 4.1, 4.3